

Bayes' Rule

- Bayesian statistics is a collection of tools that is used in a special form of statistical inference which applies in the analysis of experimental data in many practical applications in science and engineering.
- A generalization of the case where the sample space is partitioned into k subsets is covered by the theorem of total probability or the rule of elimination.
- If the events $B_1, \dots, B_k$ constitute the partition of the sample space such that $P(B_i) \neq 0$ for $i = 1, \dots, k$, then for any event A of S ,

$$P(A) = \sum_{i=1}^k P(B_i \cap A) = \sum_{i=1}^k P(B_i) P(A|B_i)$$

- Bayes Rule -

$$P(B_i | A) = \frac{P(B_i \cap A)}{\sum_{i=1}^k P(B_i \cap A)}$$

$$= \frac{P(B_{a_i}) P(A|B_{a_i})}{\sum_{i=1}^k P(B_i) P(A|B_i)}$$

$$a_i = 1, \dots, k$$

or simply put,

$$P(B_{a_i}|A) = \frac{P(B_{a_i} \cap A)}{P(A)}$$